From pieces to pipeline: Why AI isn't ready for end-to-end systematic review and meta-analysis

Abstract

Systematic reviews (SRs) are critical to evidence-based medicine. Artificial intelligence (AI), particularly large language models (LLMs), is increasingly used to answer medical questions by retrieving and citing relevant studies, as well as to support the conduct of SRs and meta-analyses (MAs). This convergence raises the prospect of answering clinical questions through increasingly automated evidence synthesis. Existing AI-assisted SR research has concentrated on discrete tasks such as literature searching, study screening, and data extraction, while recent agentic systems increasingly claim broader or end-to-end capabilities. In this Perspective, we argue that strong performance on individual tasks is insufficient to establish the validity of an automated SR workflow. Critical but understudied challenges include deciding what is worth reviewing, establishing review-level population, intervention, comparator, outcome, and study-design (PICO-S) definitions, defining extraction specifications and aligning outcomes, integrating evidence across sources, assessing risk of bias, and understanding how errors propagate through the pipeline. Evaluation should therefore move beyond stagewise accuracy toward the validity and robustness of final evidence syntheses, while accounting for rapidly evolving models and clearly specifying where human involvement enters the workflow. Current LLMs can substantially assist evidence synthesis, but reliable autonomous systematic review remains an unfulfilled goal.

Keywords: Evidence synthesis automation, Large language models, Agentic AI, Systematic reviews, Meta-analysis, Medical question answering

1 Introduction

Systematic reviews (SRs) occupy a prominent position in evidence hierarchies and are widely regarded as a cornerstone of evidence-based medicine [1–3]. Their central importance comes from their ability to systematically identify, appraise, and synthesize relevant primary evidence through transparent and reproducible methods [4, 5]. By bringing together findings across studies, SRs can provide a more complete and less biased picture of the evidence than any single study. When quantitative synthesis is appropriate, meta-analysis (MA) can pool effect estimates across diverse populations, settings, and research groups, increasing precision and potentially enhancing the generalizability of the findings [6, 7]. Accordingly, SRs provide a foundational evidence base for informing clinical practice guidelines, guiding health policy, and answering clinical questions [8–11].

Concurrently, artificial intelligence (AI)—driven largely by advances in large language models (LLMs)—is gaining traction across medicine, from clinical decision support and patient communication to documentation and question answering [12, 13]. In clinical question answering, LLM-based systems can search the literature and generate responses backed by citations. Yet a cited answer does not guarantee a valid evidence synthesis. Retrieved sources may represent only a selective subset of the relevant literature, and evaluations frequently reveal problems with citation accuracy, faithfulness, relevance, and comprehensiveness [14–17]. This gap raises a broader question: can AI reliably carry out the systematic review process end to end?

Interest in automating SRs and MAs long predates modern LLMs (e.g., [18, 19]), but recent technological advances have accelerated this effort substantially. Early efforts focused on automating discrete steps such as literature searching, screening, and data extraction (see reviews [20–27]). Recent work has demonstrated the scalability of such approaches; for example, Chen et al. used GPT-5 to assist screening, evidence classification, and metadata extraction in a systematic review of thousands of studies [28]. More recently, research has shifted toward building autonomous "agents" capable of executing an entire SR pipeline with minimal human input [29–31]. This momentum extends into commercial and institutional ecosystems: established systematic review platforms (e.g., Covidence, Rayyan) and newer startups (e.g., Elicit) now feature AI-assisted workflows, while major foundation models like OpenAI's ChatGPT (Deep

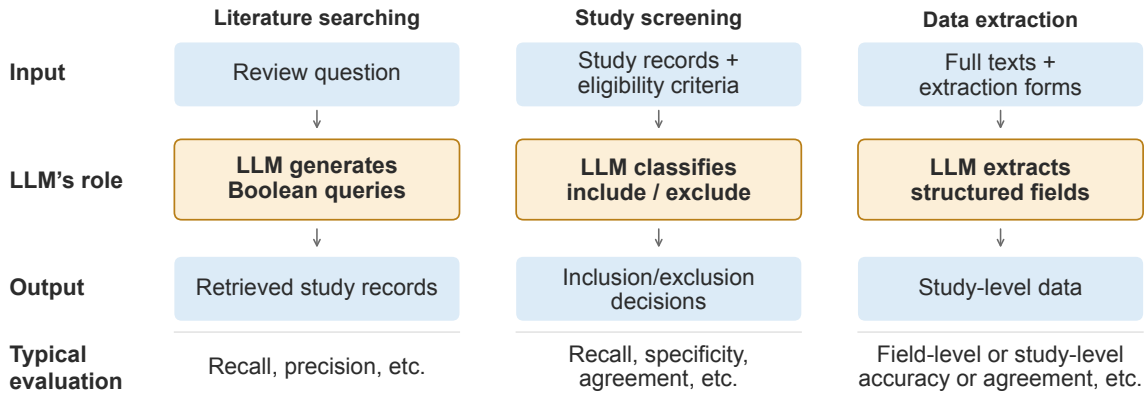


Fig. 1 Dominant tasks in LLM-assisted systematic reviews. Existing work has focused primarily on using LLMs to generate literature-search queries, classify study records during title/abstract or full-text screening, and extract structured information from included studies.

Research mode or Agent mode) and Anthropic's Claude offer direct capabilities for systematic review [32–34]. In parallel, organizations like the National Institute for Health and Care Excellence (NICE) and the National Academies have begun to establish expectations for responsible AI use [35, 36]. Yet, despite this rapid proliferation of research and tooling, actual adoption in published evidence syntheses remains limited and inconsistently reported [37, 38].

In this perspective, we argue that current AI systems are not yet ready to conduct end-to-end systematic reviews and meta-analyses. We provide a brief overview of the current landscape of AI-assisted automation for SR and MA, but focus primarily on tasks that are critical to the SR process yet remain understudied. We write with two audiences in mind. For systematic reviewers, healthcare professionals, and others considering AI-assisted tools, we hope to clarify what can reasonably be expected from current systems and where caution is still warranted. For AI researchers, we hope to highlight where effort has been overinvested and redirect attention toward these underexplored tasks.

2 Current landscape of AI-assisted systematic review

2.1 Searching, screening and extraction dominate current research

Research on LLM-assisted systematic reviews has concentrated heavily on three stages: literature searching, study screening (at the title/abstract or full-text level), and data extraction [20, 26, 34]. Figure 1 summarizes the typical roles of LLMs at each stage. In literature searching, LLMs have primarily been used as query-formulation assistants rather than as retrieval systems in their own right. Given a review question, the model identifies the main searchable concepts, proposes free-text synonyms, and generates Boolean queries for conventional bibliographic databases [39–42]. The generated queries are generally evaluated on benchmark collections derived from previously completed reviews by measuring recall/sensitivity, precision, and F-scores. Methodological contributions have focused on developing and comparing prompting strategies for query generation.

In study screening, the dominant paradigm treats the LLM as a classifier: the model reads each study alongside the eligibility criteria and produces an inclusion or exclusion decision. Methodological contributions focus on model configuration and prompt design [43–49]: zero/few shot prompting, chain-of-thought reasoning, population, intervention, comparator, and outcome (PICO)-structured templates, decision thresholds, or ensembles of prompts or models. These systems are typically evaluated by treating the inclusion decisions from a completed human review as the reference standard and measuring accuracy, recall, or specificity.

LLM-assisted data extraction leverages the zero-shot information-extraction capabilities of LLMs: study materials (e.g., full-text PDFs) can be provided to a model together with a predefined extraction form specifying fields such as study characteristics, arm-level sample sizes, outcomes of interest, and, where applicable, the relevant follow-up time and measurement scales (e.g., [50–54], see review [55]). The model then populates these fields for each study. Evaluations typically compare extracted elements against human-curated or adjudicated reference standards using measures such as accuracy or agreement, often supplemented by omission and hallucination rates. Reported performance varies widely with both extraction target and the granularity of evaluation [55]: LLMs generally perform better on explicitly

reported categorical or descriptive fields than on numerical, derived, or complex outcome data, and data-point-level evaluations tend to yield higher accuracy than complete outcome- or study-level extraction. Prompt design can further affect performance, although reported gains are generally modest and may depend on how task-specific the prompts are [53, 54].

2.2 Emerging agentic systems and end-to-end pipelines

A growing body of research has moved beyond individual LLM-based tasks toward agentic systems or integrated pipelines for conducting systematic reviews and meta-analyses [29–31, 56–64], with several representative systems compared in Figure 2. Note that some systems describe themselves as end-to-end [64], even though they implement only a subset of the tasks required for a complete SR workflow. We focus on peer-reviewed work and preprints intended for peer-reviewed publication. We do not include AI features embedded within conventional SR software platforms, such as Covidence. Most included systems focus on biomedical applications, although we also consider relevant work in other fields (e.g., psychology [62], environmental science [64]).

TrialMind [56] and Otto-SR [29] are the more comprehensive examples of these systems. Both have been applied across multiple clinical domains and incorporated human evaluation of system performance. The majority of the systems still focus on the screening and data-extraction stages. Some extend further upstream by using LLMs to develop literature search strategies [56, 59], while others extend downstream by feeding extracted data into statistical pooling and meta-analysis [60]. Many are also limited to specific domains [59, 60], raising questions about whether they can generalize across review questions like professional systematic reviewers.

A notable feature of many such systems is that, despite being presented as end-to-end pipelines, their evaluation often remains stagewise (but see [57], which evaluates extraction through agreement in final meta-analytic conclusions). Individual stages are commonly evaluated using inputs derived from the original human-conducted review rather than outputs generated by preceding stages of the pipeline. For example, screening may be evaluated using the set of studies retrieved by human reviewers, while data extraction may be evaluated using studies retained by human reviewers together with predefined extraction forms. This evaluation strategy effectively isolates the performance of individual components but, in practice, makes the assessment of these agentic systems conceptually similar to earlier task-specific evaluations of LLMs.

3 The missing pieces in end-to-end SRs and MAs

3.1 Deciding what is worth reviewing

An SR is a question-driven enterprise. The review question shapes the eligibility criteria, search strategy, outcomes and analysis, and is specified prospectively in a protocol before the review is conducted. A review question typically arises from an important gap in current knowledge or from the accumulation of new primary studies that make a synthesis worthwhile [65, 66]. Formulating the right question is therefore critical: a flawlessly conducted review of a redundant or unimportant question consumes substantial resources while adding little to the evidence base [67, 68].

Given the importance of question formulation, one might expect AI to help here, given its capacity to search and synthesize how review questions have been proposed, how clinical needs are shifting, and what new primary studies report. In practice, however, current "end-to-end" systems do not attempt to decide what is worth reviewing. Otto-SR, for instance, takes a protocol (objectives and eligibility criteria) together with search results as its inputs [29]. TrialMind goes a step earlier by generating the search query and eligibility criteria, but still begins from a human-defined PICO [56]. Other systems are similar in this regard (see Figure 2). Consistent with this pattern, a scoping review of LLM applications [20] found only a handful of studies touching question framing, all exploratory [69, 70], with similar exploratory work appearing more recently [34]. This limited attention is understandable: starting from a fixed question allows these systems to be evaluated against existing reviews, whereas deciding which questions are worth reviewing is inherently stakeholder-dependent and lacks a single gold standard [71, 72]. More broadly, evidence on machine ideation is cautionary: when LLMs were asked to generate novel research ideas in NLP, the ideas initially appeared more novel than human ones, but that advantage collapsed once the ideas were actually executed [73].

System	Search / retrieval	Screening	Extraction	Risk of bias	Statistical pooling	Evaluation scope and boundary
TrialMind (Wang, 2025)	Y	Y	Y	--	H	100 clinical SRs / 2,220 studies; numerical standardization feeds human-run R pooling
otto-SR (Cao, preprint)	H	Y	Y	Y	H	Task benchmarks and reproduction/update of 12 Cochrane reviews; search and analysis were investigator-performed using inherited specifications
A4SLR (Lee, 2025)	Y	Y	Y	Y	--	Two clinical use cases: lung cancer and perinatal mood/anxiety disorders
AgentSLR (Padarha, 2026)	Y	Y	Y	--	--	Nine priority pathogens; structured epidemiological extraction and descriptive/narrative reports
LUMEN (Huang, preprint)	Y	Y	Y	Y	Y	Five clinical review applications plus two screening benchmarks; expert analytical decisions remain
AutoSynthesis (Taherinezhad, preprint)	Y	Y	Y	Y	Y	Demonstration on LLM persuasion, compared with published meta-analysis; limited breadth of validation
MetaMind (Livieratos, 2026)	Y	H	Y	--	Y	Ulcerative colitis / Crohn's disease NMA; manual feasibility, eligibility, and model-fit assessment
LatteReview (Rouzrokh, preprint)	--	Y	Y	--	--	Modular screening / extraction framework; framework examples and task evaluations, not full SR validation
AutoMetaCoder (Min, preprint)	--	Y	Y	--	--	Screening implemented but not formally evaluated; extraction evaluated on 55 studies from a psychology meta-analysis
InsightAgent (Qiu, 2025)	--	Y	--	--	--	Interactive abstract-level screening and narrative synthesis; no structured data extraction; nine-professional user study
Manalyzer (Xu, preprint)	Y	Y	Y	--	--	729 papers across atmosphere, agriculture, environment; generates general analyses rather than conventional effect-size pooling
meta-pipe (Lin, preprint)	Y	Y	Y	Y	Y	Implemented architecture; no empirical validation data; five mandatory human decision points

Y Implemented by the system H Human-performed or inherited from the source review -- Not implemented

Fig. 2 Overview of LLM-based and agentic systems supporting systematic review and meta-analysis. Systems are compared across literature search and retrieval, screening, data extraction, risk-of-bias assessment, and statistical pooling. Statistical pooling refers to combining study-level effect estimates rather than narrative synthesis or general data analysis. The final column summarizes evaluation settings and key limitations.

3.2 Establishing review-level PICO-S

The essence of an SR and MA is to synthesize information from individual primary studies into a unified analysis, which requires the non-trivial task of defining review-level population, intervention, comparator, outcome, and study-design (PICO-S) criteria. These review-level definitions directly determine eligibility criteria and can therefore substantially shape the resulting synthesis. Yet this step is largely overlooked in current LLM-based SR research, which typically begins from human-specified review questions, PICO elements, or eligibility criteria, rather than examining how they are formulated [20, 29, 56] (but see the exploratory study by Chen et al. [74]).

A classic challenge in defining review-level PICO-S is the lumper–splitter problem. Consider the intervention (I): interventions relevant to the same review question are typically similar enough to be considered together, while still differing in potentially meaningful ways. A lumper emphasizes these shared features and favors grouping similar interventions for pooled analysis, whereas a splitter emphasizes their differences and favors analyzing distinct interventions separately. For example, a review of newer-generation antidepressants for depression may treat fluoxetine, paroxetine, sertraline, and other drugs as a single intervention class (lumper), or analyze the effects of each compound separately (splitter) [75].

Another key consideration is how the intervention itself is defined, for example, whether studies that deliver it alongside a co-intervention are eligible for inclusion. Consider two scenarios. In the first, an

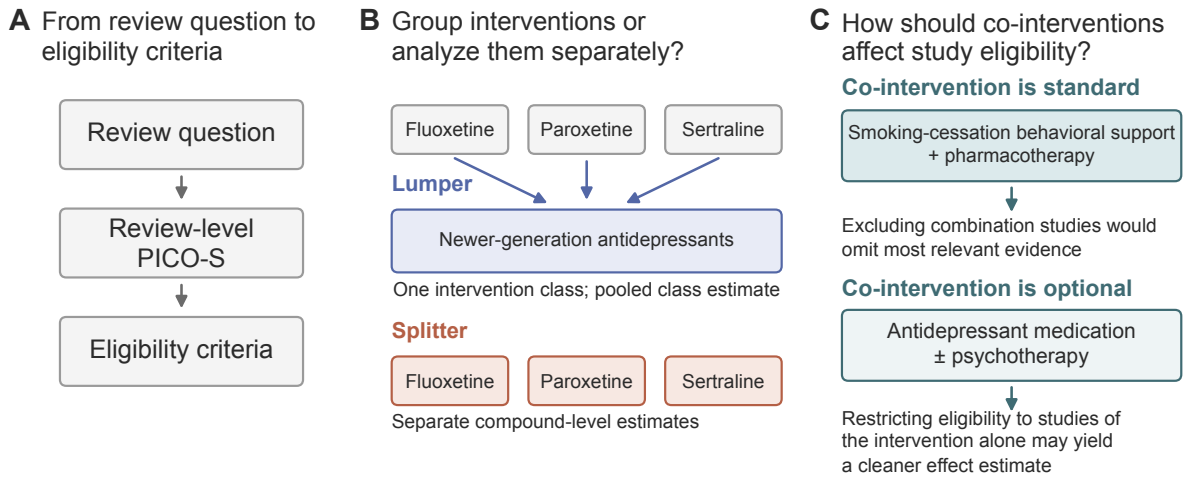


Fig. 3 The critical step of establishing review-level PICO-S. A. Reviewers translate a review question into review-level population, intervention, comparator, outcome, and study-design definitions, which determine study eligibility. This formulation step is typically supplied by humans in current LLM-based systematic-review workflows. **B.** The lumper–splitter decision determines whether related compounds—illustrated by fluoxetine, paroxetine, and sertraline—are grouped as one intervention class (newer-generation antidepressants) and pooled or analyzed separately to obtain compound-level estimates. **C.** Eligibility rules for co-interventions should reflect clinical context. When a co-intervention is standard, as with smoking-cessation behavioral support delivered with pharmacotherapy, excluding combination studies may omit most relevant evidence. When a co-intervention is optional, as with psychotherapy added to antidepressant medication, restricting eligibility to studies of the intervention alone may yield a cleaner effect estimate. These examples illustrate why reviewers should pre-specify co-intervention rules based on real-world clinical practice.

intervention is almost always delivered with a co-intervention; for example, behavioral support for smoking cessation is typically evaluated as an adjunct to pharmacotherapy. Requiring the intervention to be studied in isolation would exclude most relevant trials and yield estimates lacking real-world applicability. In the second scenario, the co-intervention is optional, but its presence may alter the treatment effect through interaction, such as when psychotherapy is added to antidepressant medications. Therefore, there is no universal rule; reviewers must pre-specify their approach to co-interventions based on real-world clinical practice.

A comparable decision process applies when defining the comparator (C). Labels such as "placebo", "waitlist", "no treatment", and "usual care" can vary significantly in meaning across studies and settings [76]. For example, "usual care" for managing hypertension might involve routine pharmacological treatment plus lifestyle counseling in one healthcare system, but a less intensive treatment in another [76, 77]. Pooling such comparators without accounting for these differences can obscure meaningful heterogeneity and yield misleading effect estimates. Reviewers must therefore decide what specific comparators are appropriate for their review and determine whether different control conditions should be grouped together or analyzed separately.

Reviewers face similar boundary decisions for population (P) and study design (S) criteria. For populations, they must decide whether patient differences, such as age, disease severity, diagnostic criteria, or comorbidities, are distinct enough to separate or similar enough to combine. For study design, reviewers must determine which evidence types (e.g., randomized vs. observational) and specific features (e.g., cluster or crossover designs) are acceptable based on the specific review question. Corresponding challenges apply to outcomes (O), which we discuss in detail in Section 3.3.

3.3 Defining extraction specifications and aligning outcomes

Designing well-specified data extraction forms is an important step in data extraction. Existing LLM-based data extraction studies typically start with a predefined extraction form or schema that specifies the information to be extracted (e.g., [29, 59]). These specifications determine not only what information is collected but also how choices among multiple eligible outcome measures, time points, analysis populations, and effect estimates are resolved [78, 79]. Clear outcome definitions are therefore essential for accurate data extraction, whether performed by human reviewers or LLMs. In a large-scale reproducibility study of human extraction of adverse-event data, 29.9% of unreproducible trial-level extractions were classified as ambiguous, primarily because outcome definitions were unclear [80]. LLM extraction appears similarly sensitive to such complexity. Performance deteriorated when outcomes were ambiguous, multiple similar measures were reported, or the required values had to be inferred [81].

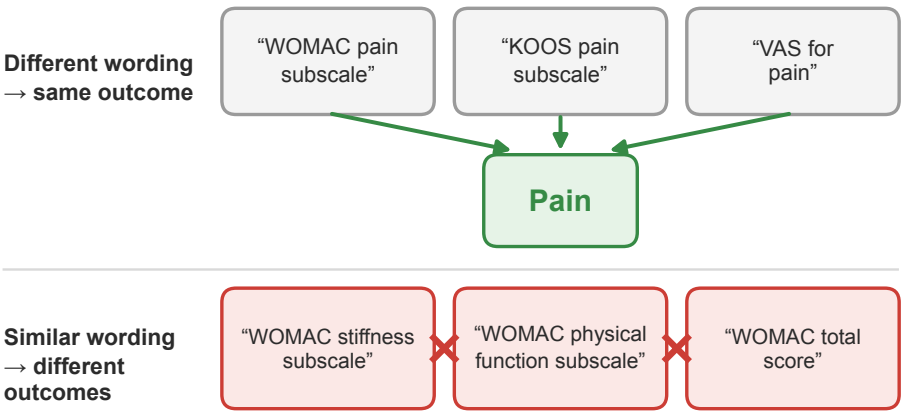


Fig. 4 Illustration of outcome mapping in a systematic review. Outcome measures with different names may represent the same outcome. For example, in a Cochrane review of exercise for knee osteoarthritis [88], pain was assessed using the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) pain subscale [89, 90], the Knee Injury and Osteoarthritis Outcome Score (KOOS) pain subscale [91, 92], and a visual analogue scale (VAS) [93, 94]. These measures can all be mapped to the common outcome of pain. Conversely, the WOMAC stiffness subscale, WOMAC physical function subscale, and WOMAC total score represent different outcomes and should not be combined, despite sharing the WOMAC acronym.

The lumper-splitter problem discussed in Section 3.2 also arises when defining review-level outcomes [82]: reviewers must decide which outcomes to include and how to align them across studies. Outcomes with similar labels but inherently different meanings must be kept separate, whereas conceptually similar outcomes with different labels must be mapped onto a common construct (see Figure 4 for examples). Failing at the first task yields what psychological science calls the jingle fallacy; failing at the second, the jangle fallacy [83–85]. Such problems have received far less attention in biomedicine, but a growing number of biomedical studies have started to document outcome-reporting heterogeneity [86, 87].

Currently, no single methodology can be considered universally correct for mapping these outcomes, so reviewers must navigate this process flexibly, guided by practical constraints, specific research goals, and disciplinary traditions [82]. For instance, one review might synthesize data across various follow-up periods into a single outcome construct, whereas another might treat those periods as distinct outcomes. Regardless of the chosen approach, reviewers should explicitly justify their decisions to avoid excessive lumping or splitting of the data.

Whether AI can make reasonable decisions when selecting and aligning outcomes and specifying relevant extraction rules remains an open question that has received little attention (but see exploratory work in [22, 95]), despite its potential impact on the quality of evidence synthesis. Greater standardization in clinical research, such as through core outcome sets and standardized outcome definitions, measurement instruments, and follow-up time points, could make this task more tractable by reducing ambiguity in mapping outcomes across studies [96]. One prominent effort is the Core Outcome Measures in Effectiveness Trials (COMET) Initiative, which promotes the development and use of standardized core outcome sets for specific health conditions and research areas [97–99].

3.4 Characterizing error propagation

Although many studies have reported strong LLM performance on individual SR tasks, evaluating each stage in isolation raises two important concerns (Figure 5). First, errors can propagate through the SR pipeline: mistakes made at an earlier stage may carry forward and become more consequential if they are not detected or corrected downstream [80, 100]. Human reviewers may sometimes catch such errors at later stages and revisit earlier decisions; for example, Cochrane guidance notes that data extraction can provide an additional opportunity to confirm study eligibility when relevant eligibility information is not apparent during screening [101]. In contrast, stagewise AI systems may simply accept upstream outputs as given unless cross-stage checking is explicitly built into the workflow [29].

Second, not all stage-level errors carry the same weight in the final evidence synthesis. For example, mistakenly including an ineligible commentary article may have little downstream impact if it contains no extractable data. In contrast, accidentally including a study with an ineligible intervention that contributes outcome data could directly distort the pooled evidence and potentially bias conclusions. Therefore, reporting high overall accuracy alone may not be very informative. It may be useful to also evaluate and report AI performance on consequential edge cases, which may account for only a small proportion

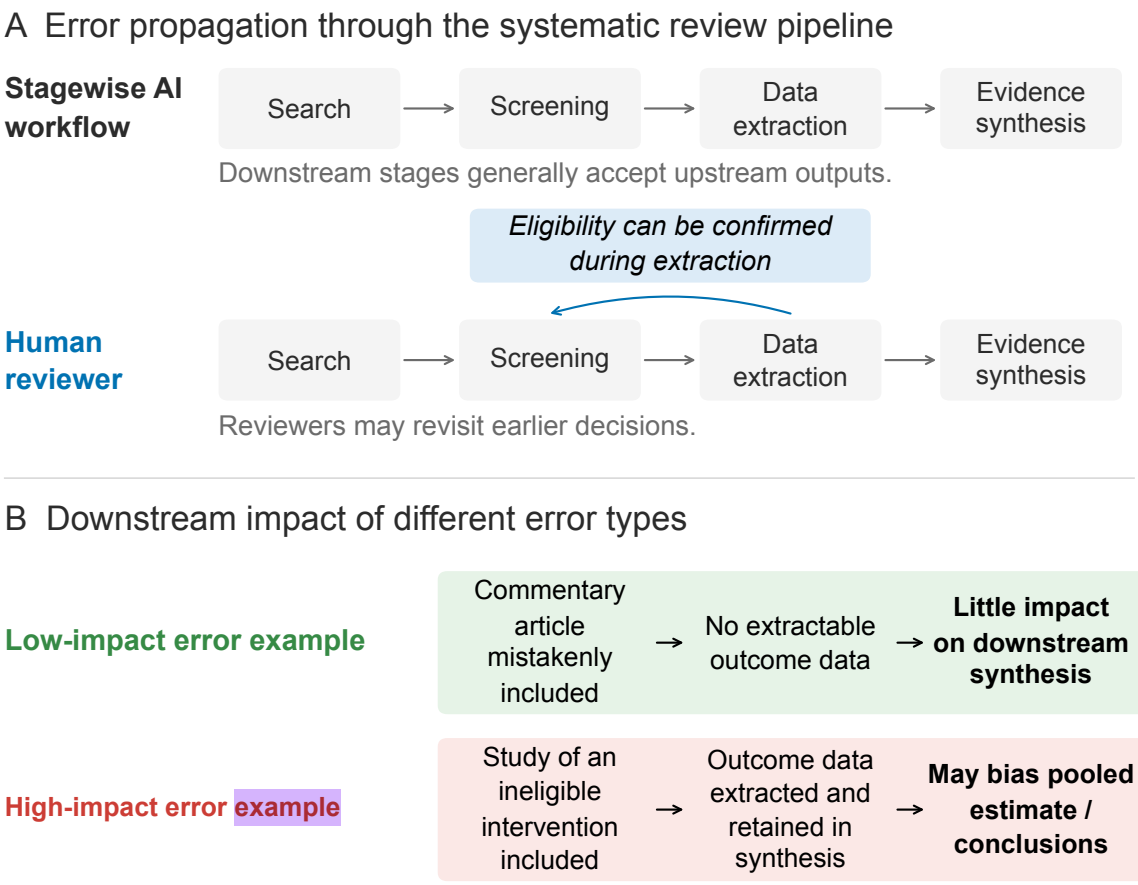


Fig. 5 Error propagation through the systematic review pipeline. A. Stagewise AI workflows may carry upstream errors forward unless cross-stage checks are explicitly implemented. Human review is less strictly stagewise: information obtained at later stages can prompt reassessment of earlier decisions. For example, information encountered during data extraction may confirm study eligibility or lead reviewers to revise an earlier screening decision. **B.** Example of downstream impacts of different error types. For screening errors, a commentary article without extractable outcome data may have little impact on synthesis, whereas a study of an ineligible intervention that contributes outcome data may bias the pooled estimate or review conclusions.

of errors but have a disproportionate impact on the final synthesis. Such errors are easily obscured by aggregate accuracy metrics.

3.5 Integrating evidence from multiple data sources

The data required for extraction may be distributed across multiple sources: the main text, tables, and figures of the primary article, or its supplementary materials [100]. Outcomes with nonsignificant findings are particularly vulnerable to incomplete or selective reporting, such that the data needed for evidence synthesis may not be available in the main publication [102, 103]. Such nonsignificant outcomes may nonetheless be essential to an unbiased synthesis and should not be overlooked. In designing an end-to-end AI system, it is hard to ensure that those supplementary sources are properly retrieved and searched. This problem is compounded by the growth in supplementary materials, which has increased substantially in both number and length [104, 105]. Human reviewers can actively search across these sources when required, whereas an automated system may struggle to distinguish information that is genuinely absent from information that was simply missed during retrieval, potentially leading to hallucinated outputs [106].

A related challenge is that human reviewers commonly contact the authors of primary studies to confirm ambiguous values or obtain data absent from the publication [107, 108], although this need may be reduced by improved adherence to reporting standards such as CONSORT (Consolidated Standards of Reporting Trials) [109, 110]. An AI agent could technically initiate such correspondence, but whether it should is less clear: these exchanges rest partly on the professional identity and credibility of the requesting researcher [111], which do not transfer straightforwardly to an automated system. Meanwhile, widespread automated outreach could overwhelm study authors with requests [112].

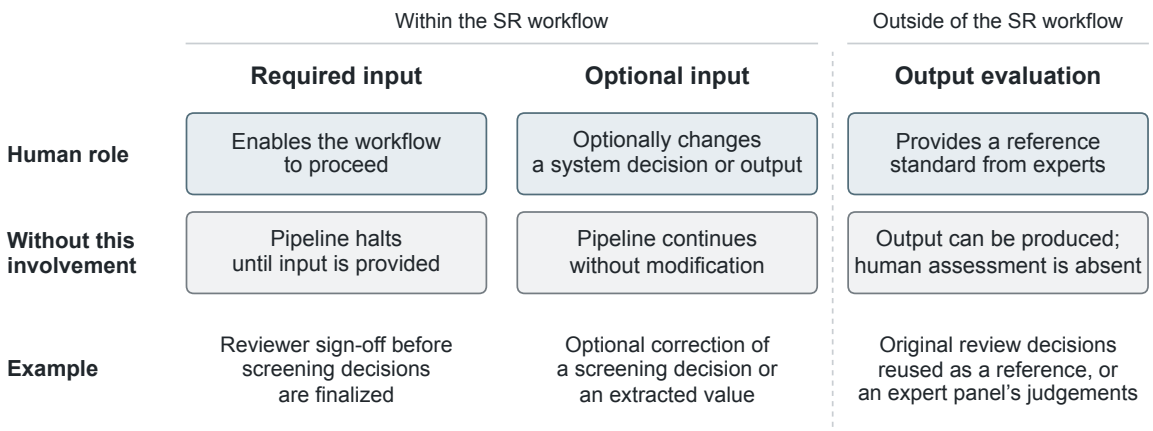


Fig. 6 Distinguishing human roles in AI-assisted systematic reviews. Within the systematic review workflow, required human input enables the workflow to proceed and causes the pipeline to halt until the input is provided, whereas optional input may modify a system decision or output but is not necessary for the workflow to continue. Outside the workflow, human involvement may instead serve to evaluate system outputs by providing expert judgements or reference decisions.

3.6 Assessing risk of bias reliably

Risk of bias (RoB) assessment is an essential component of the SR and MA pipeline [113], even though it may appear to be just one downstream step. Without a high-quality RoB assessment, the reliability of the resulting forest plot becomes difficult to assess, since a high-risk-of-bias study can contaminate the aggregated effect size. In human-conducted SRs, researchers can address this by running a sensitivity analysis that excludes high-risk studies to test the robustness of the aggregated effect.

Research specifically examining AI-based RoB assessment exists but remains limited given the wide range of RoB tools [113, 114]. Empirical evaluations of AI-based RoB assessment have found highly mixed agreement with human judgments, ranging from slight to moderate to high agreement [115–120]. Moreover, the effect of inaccurate AI-based RoB assessments on the acceptance or interpretation of aggregated effect sizes in forest plots remains understudied in existing end-to-end SR and MA systems [29–31]. This lack of established reliability creates a concerning blind spot: users of AI-automated SR and MA systems may place undue confidence in forest plots without recognizing that their conclusions can be undermined by errors in the underlying RoB assessments.

3.7 Keeping pace with evolving proprietary LLMs

Performance across stages of an AI-assisted systematic review depends substantially on the capabilities of the underlying LLM [26]. Model performance also appears to be task dependent: a model that excels at one stage may not necessarily be optimal at another. However, existing evidence provides limited guidance for selecting the most appropriate model for each task. Comparisons across published studies are largely indirect because evaluations differ in the models and model versions tested, prompts, datasets, reference standards, and performance metrics [20, 26]. Although multi-stage benchmarks such as TrialReviewBench have begun to emerge [56], the field still lacks a widely adopted and continuously maintained benchmark suite that enables consistent comparisons across review tasks, clinical domains, and successive model generations.

This problem is compounded by the rapid and often opaque evolution of proprietary LLMs [121, 122]. Model behavior can change substantially across releases and generations [123, 124], sometimes altering which prompting strategy performs best [125]. Consequently, prompts and pipelines validated with one model version cannot be assumed to retain their performance after subsequent updates.

3.8 Distinguishing the roles humans play

The majority of the AI-based SR systems still rely on human input at one or more stages of the workflow [21]. Human-AI interaction has well-documented failure modes, including complacency and automation bias [126]. Moreover, a recent meta-analysis found that human-AI collaboration, on average, performed worse than the best of human or AI alone especially in tasks that required making decisions [127]. Similar automation-related failure modes may arise in AI-automated SR workflows. For example, when AI

takes over much of the screening process, human reviewers may have less exposure to individual studies and therefore fewer opportunities to notice nuances or contextual information that could be valuable for downstream data extraction or interpretation. Thus, rather than simply aiming to minimize human involvement, AI-automated SRs should be designed to consider where humans should sit in the system, and also state clearly what tasks humans need to do and how to do that to work with the AI assistance.

In existing systems, the different roles humans play are not always clearly distinguished between the following: stages where human input is *required* and stages where it is merely *permitted* (see Figure 6). A required input is one without which the system cannot produce an output at all; it halts the pipeline until a reviewer signs off. A permitted modification is one the system does not need but will accept, such as an optional correction to a screening decision or an edited extraction. A separate case is human involvement in evaluating the system's output. The human contribution here may be the original review authors' decisions, reused as a reference standard, or the judgements of an expert panel convened specifically for the evaluation. These types of human involvement should not be mistaken for the system's own performance.

4 Future directions

We propose four future directions for advancing AI-assisted systematic reviews.

First, research should expand to less-studied review tasks that are more subjective and currently more human involved, including question selection, PICO-S formulation, and outcome alignment. Future studies should examine how AI can support the development of answerable and clinically meaningful questions, translate these questions into clear eligibility criteria, and align outcomes across studies while preserving clinically meaningful distinctions.

Second, evaluation and benchmarking should encompass both individual tasks and the complete review pipeline. We propose developing and regularly updating benchmarks that link review questions to included studies, extracted data, and pooled results. These benchmarks should incorporate challenging cases, including studies that are easily missed or misclassified, and enable errors to be traced across review stages. Evaluations should document model versions and system configurations and assess whether the complete pipeline preserves the validity and robustness of the final synthesis.

Third, system design should be informed by empirical comparisons of the modes and extent of human involvement. Comparative studies should examine how independent verification, targeted review of uncertain outputs, and intervention at predefined decision points affect accuracy, efficiency, and error correction. Human–AI interaction should be an explicit component of system evaluation, with clear descriptions of where and how reviewers intervene, what information they receive, and how disagreements are resolved.

Fourth, improvements in data reporting and cleaning should complement advances in system design. These efforts should promote consistency across abstracts, full-text articles, and supplementary materials. We propose standardizing reporting and developing structured databases that serve as auditable "mirrors" of original publications, providing traceable inputs to AI systems. This approach could streamline data preparation while supporting transparency and reproducibility.

5 Conclusion

LLMs show strong potential for assisting systematic reviews (SRs) and meta-analyses (MAs), but critical gaps remain before end-to-end workflows become feasible. Here, we surveyed the current landscape of LLM-based SR and MA research and found that most work focuses on literature searching, study screening, and data extraction. Even systems described as end-to-end often remain largely modular in nature and omit important components of the review process. We therefore highlight several underexplored challenges, including deciding what questions are worth reviewing, defining review-level PICO-S, selecting and aligning outcomes across studies, integrating information from multiple sources, and conducting risk-of-bias assessment. Future evaluations should also move beyond stagewise performance metrics to examine how system-level errors affect the quality and conclusions of the final review. In addition, researchers should account for the proprietary and rapidly evolving nature of LLMs when designing evaluations so that findings remain interpretable and useful over time. Finally, systems should clearly specify where and how human involvement enters the workflow. Addressing these issues will be essential before end-to-end systematic review and meta-analysis can become a credible goal.

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